stmunits package description

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These are the unit definitions for `stmlatex`. It is build upon the siunitx package.

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1. Usage

1.1. Load the whole `stmunits` package

1.1.1. Description

This is an interface package which loads the base package and the style choice.
By default the package loads

- `stmunitsbase.sty`
- `stmunitsitems.sty`
- `stmunitsstyles.sty`

See subsubsection 1.1.2 for options to change the default package behavior.

1.1.2. Options

Option *styles* This is a boolean option. Expected values are either `true` or `false`. It controls whether to load the predefined styles.

```
\usepackage[styles=true|false]{stmunits}
```

`styles=true` is the default. It is used in case `styles=false` is not set explicitly.

Option *style* This option value is the equivalent to section 1.4.2 and is passed through to the underlying package.

1.2. `stmunitsbase`

This package loads the underlying base package.

1.3. `stmunitsitems`

This loads new unit definitions, not initially defined in the base package. See section 2 for the definitions.

1.4. `stmunitsstyles`

1.4.1. Description

This package contains styles for the representation of units and numbers.

1.4.2. Options

Option *style* This is a string option. A list of possible values is shown in section 3.

```
\usepackage[style=stmdefault]{stmunitsstyles}
```

`style=stmdefault` is the default. It is used in case no other style is set explicitly.

2. Items

Table 1: Additional unit definitions

Length			
`\si {\foot }`	ft	`\si {\inch }`	in
Temperature			
`\si {\degreeFahrenheit }`	°F	`\si {\degreeRankine }`	°R
Force			
`\si {\dyn }`	dyn	`\si {\poundforce }`	lbf
Pressure			
`\si {\barye }`	Ba	`\si {\psi }`	psi
Density			
`\si {\slug }`	slug		
Energy			
`\si {\erg }`	erg		
Misc.			
`\si {\promille }`			

3. Styles

There are different predefined styles available to format your numbers and units.

3.1. `default`

This loads the default style from the underlying base package and does not apply individual styles.

3.2. `stmdefault`

This loads unit definition style for the stm package. Have a look at the code in section A.4 for details.

A. The code

A.1. stmunits.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes unit definitions.
% Based upon the siunitx package:
%   https://ctan.org/pkg/siunitx
%
% Usage
%  - Preamble:
%    - \usepackage{stmunits}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmunits}[2023/02/12 STMs custom LaTeX unit definitions]

% ---------------------------
% Options
% ---------------------------

% For options
```

```
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

% Option group
\SetupKeyvalOptions{%
  family=stmunits,%
  prefix=stmunits@,%
  setkeys=\kvsetkeys,%
}

% Items
\DeclareBoolOption[true]{items}

% Load styles
\DeclareBoolOption[true]{styles}

% Apply style name
\DeclareStringOption[stmdefault]{style}

% Process options
\ProcessKeyvalOptions{stmunits}

% ----------------------------
% Modules
% ----------------------------

% Load the base package
\@ifpackageloaded{stmunitsbase}{}{%
  \RequirePackage{stmunitsbase}%
}%

% Load the styles
\ifstmunits@styles
  \@ifpackageloaded{stmunitsstyles}{}{%
    \RequirePackage[
      style=\stmunits@style
    ]{stmunitsstyles}%
  }%
\fi

% Load the items
\ifstmunits@items
  \@ifpackageloaded{stmunitsitems}{}{%
    \RequirePackage{stmunitsitems}%
  }%
```

```
\fi

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.2. stmunitsbase.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes unit definitions.
% Based upon the siunitx package:
%   https://ctan.org/pkg/siunitx
%
% Usage
%  - Preamble:
%    - \usepackage{stmunitsbase}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                  __/|__
%                                 /_/_/_/
%            www.dlr.de/sy/en       |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
```

```
    comment about what it's used for
33 \ProvidesPackage{stmunitsbase}[2023/02/12 STMs custom LaTeX unit
    definitions]
34
35 % --------------------------
36 % Package
37 % --------------------------
38
39 \@ifpackageloaded{siunitx}{}{%
40   \RequirePackage{siunitx}
41 }%
42
43 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
44 % That's it                        %
45 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
46
47 % Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
48 \endinput
```

A.3. stmunitsitems.sty

```
1  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
2  % Header                          %
3  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
4  %
5  % This file includes unit definitions.
6  % Based upon the siunitx package:
7  %   https://ctan.org/pkg/siunitx
8  %
9  % Usage
10 %  - Preamble:
11 %    - \usepackage{stmunits}
12 %
13 % Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
14 %                       Initial draft
15 %
16 % Contact:   Martin Raedel,  martin.raedel@dlr.de
17 %            DLR Lightweight Systems
18 %
19 %                                __/|__
20 %                               /_/_/_/
21 %            www.dlr.de/sy/en     |/ DLR
22 %
```

```
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                           %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmunitsitems}[2023/02/12 STMs custom LaTeX unit
    definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{stmunitsbase}{}{\RequirePackage{stmunitsbase}}%

% ---------------------------
% New units
% ---------------------------

% Length
\DeclareSIUnit\foot{ft}
\DeclareSIUnit\inch{in}

% Temperature
\DeclareSIUnit\degreeFahrenheit{^{\circ}F}
\DeclareSIUnit\degreeRankine{^{\circ}R}

% Force
\DeclareSIUnit\dyn{dyn}
\DeclareSIUnit\poundforce{lbf}

% Pressure
\DeclareSIUnit\barye{Ba}
\DeclareSIUnit\psi{psi}

% Density
\DeclareSIUnit\slug{slug}

% Energy
```

```
\DeclareSIUnit\erg{erg}

% Promille
\DeclareSIUnit[number-unit-product = \,]{\promille}{\textperthousand}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                      %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.4. stmunitsstyles.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes unit definitions.
% Based upon the siunitx package:
%   https://ctan.org/pkg/siunitx
%
% Usage
%  - Preamble:
%    - \usepackage{stmunitsstyles}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
```

```
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmunitsstyles}[2023/02/12 STMs custom LaTeX unit style
    definitions]

% ----------------------------
% Base package
% ----------------------------

\@ifpackageloaded{stmunits}{}{\RequirePackage{stmunits}}%

% ----------------------------
% Options
% ----------------------------

% For options
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

\SetupKeyvalOptions{%
  family=stmunitsstyles,%
  prefix=stmunitsstyles@,%
  setkeys=\kvsetkeys,%
}

% Style name
\DeclareStringOption[stmdefault]{style}

% Process options
\ProcessKeyvalOptions{stmunitsstyles}

% ----------------------------
% Package
% ----------------------------

% ----------------------------
% Setup
% ----------------------------

\sisetup{%
  exponent-product = \cdot,%
  inter-unit-product = \ensuremath{{}\cdot{}},%
  %binary-units=true,%
```

```
}

% ---------------------------
% Style
% ---------------------------

\@namedef{stmunitsstyles@style@default}{%
  % This here is empty to load the standard sisetup
}

\@namedef{stmunitsstyles@style@stmdefault}{%
  \sisetup{%
    allow-number-unit-breaks = false,%
    %binary-units             = true,%
    detect-all,%
    exponent-product         = \cdot,%
    %output-product           = \cdot,%
    inter-unit-product       = \ensuremath{{}\cdot{}},%
    separate-uncertainty     = true,%
  }
}

% Evaluate option settings
% https://tex.stackexchange.com/a/249102
\newcommand*{\stmunitsstyles@evaluate}[1]{%
  % Check, if option value in \<prefix>@<option> exists
  \@ifundefined{stmunitsstyles@#1}{%
    % Should not happen
    \PackageError{stmunitsstyles}{Evaluating unknown option `#1'}\@ehc
  }{%
    % Check, if there is an implementation for the value of the option
    % in macro \<prefix>@<option>@<value>
    \@ifundefined{stmunitsstyles@#1@\csname stmunitsstyles@#1\endcsname}{
    %
      \PackageError{stmunitsstyles}{%
        Unknown option setting: #1=%
        \csname stmunitsstyles@#1\endcsname
      }\@ehc
    }{%
      % Call the implementation for the value
      \csname stmunitsstyles@#1@%
              \csname stmunitsstyles@#1\endcsname
      \endcsname
      \relax
```

```
    }%
  }%
}

\stmunitsstyles@evaluate{style}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```